Group 4: Healthcare Analytics

Healthcare Appointment Scheduling Database

Business Scenario

The Healthcare Appointment Scheduling Database streamlines how hospitals, clinics, and medical centers manage appointments, doctor availability, medical services, and follow-up care.

It aligns patient preferences such as preferred dates, times, and treatment types with qualified and available healthcare providers.

Patients and doctors can easily access essential information including contact details, service descriptions, and treatment costs.

By integrating scheduling, communication, and service details into one system, the platform enhances patient convenience and improves overall healthcare accessibility.

Business Problems Solved

- Prevents scheduling conflicts by matching appointments to real-time doctor availability.
- Ensures patients understand treatment costs and service details before booking.
- Improves communication across patients, doctors, staff, and executives through centralized information access.
- Enables administrators to review payments, track performance, and make informed business decisions.
- Supports better post-treatment care by storing prescriptions and follow-up details.
- Reduces administrative inefficiencies, enhances patient satisfaction, and improves operational oversight in medical centers.

Relational Models

- The data is organized into independent tables for Patients, Doctors, Services, Appointments, Payments, Departments, etc.
- Primary keys are used to identify each record
  - For example: PatientID, DoctorID, AppointmentID
- Foreign keys model real world links
  - Doctors -> Departments
  - Appointments -> Patient Availability, Doctor Availability, and MedicalService
  - Payments -> Appointments
  - Prescriptions -> Appointments
- Each relationship reflects how these roles interact with Healthcare operations

Normalization

- First Normal Form (1NF)
  - All tables contain atomic values: Names, addresses, phones, emails, dates, times, and numeric amounts are stored in different columns.
  - There are no repeating groups nor multi-valued attributes; the multiple availabilities or appointments are represented by multiple rows, not by multiple columns.
- Second Normal Form (2NF)
  - 2Nf condition implies the absence of only non-full dependencies on composite keys.
  - Every table has a single-column PK
- Third Normal Form (3NF)
  - The 3NF requirement is about the absence of transitive dependencies (non-key attributes dependent on other non-key attributes).
  - The schema has been normalized to at least Third Normal Form (3NF):

5 Best Queries: Query 1

Query 1: Total Revenue Generated per Medical Service.

Insightful Business Question: What is the total revenue generated per medical service during a specific month?

Helps evaluate **monthly financial performance** of each medical service.

Uses Payment, Appointment, and MedicalService tables to calculate **total revenue** for November 2025.

Results are **sorted in descending order**, highlighting the most profitable services.

Enables executives to assess **profitability**, optimize pricing, and plan service expansion.

Identifies cases where patients did not fully pay, revealing potential **affordability concerns** and opportunities to improve patient-friendly financial policies.

```sql
-- Finding the total revenue generated from per medical
service during a specific month.
SELECT
ms.MedicalServiceName,
SUM(p.AmountPaid) AS TotalRevenue
FROM HealthcareDB.Payment p
JOIN HealthcareDB.Appointment a
ON p.AppointmentID = a.AppointmentID
JOIN HealthcareDB.MedicalService ms
ON a.MedicalServiceID = ms.MedicalServiceID
WHERE p.PaymentDate BETWEEN '2025-11-01' AND '2025-11-30'
GROUP BY ms.MedicalServiceName
ORDER BY TotalRevenue DESC;
```

Row	MedicalServiceName	TotalRevenue
1	X-Ray Imaging	100
2	General Checkup	75
3	Blood Test	50
4	Vaccination	30
5	MRI Scan	0

5 Best Queries: Query 2

Query 2 – Most frequent visitors in the last 6 months

Business Question – Which patients have visited the medical center the most in the last 6 months?

Identifies high need patients to help meet their needs

Identifies loyal customers for loyalty tracking

Orders by TotalVisists in descending order

Calculates last 6 months with date_diff

Enables quick and efficient decisions to be made in response to high volume visitors

```sql
-- Which patients have visited the medical center the most in the last 6 months.
SELECT p.PatientName, p.PatientID, COUNT(a.AppointmentID) AS TotalVisits
FROM HealthcareDB.Patient  p
JOIN HealthcareDB.PatientAvailability pav
  ON p.PatientID = pav.PatientID
JOIN HealthcareDB.Appointment a
  ON a.PatientAvailabilityID = pav.PatientAvailabilityID
WHERE DATE_DIFF(CURRENT_DATE(), a.AppointmentDate, MONTH) <= 6
GROUP BY p.PatientID, p.PatientName
ORDER BY TotalVisits DESC;
```

Row	PatientName	PatientID	TotalVisits
1	Sarah Jones	1	1
2	Priya Gupta	3	1
3	Daniel Connors	4	1
4	Emily Rivera	5	1
5	Michael Lin	2	1

5 Best Queries: Query 3

```
Finding the percentage of appointments that are fully paid, partially
paid and unpaid.

SELECT

PaymentStatus,

COUNT(*) AS StatusCount,

ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM HealthcareDB.Payment),
2) AS Percentage FROM HealthcareDB.Payment

GROUP BY PaymentStatus

ORDER BY Percentage DESC;
```

Row	PaymentStatus	StatusCount	Percentage
1	Paid	3	60.0
2	Partial	1	20.0
3	Pending	1	20.0

Query 3

- Percentage of appointments that are fully paid, partially paid, or unpaid

Key Findings

- 60 % appointments paid
- 20% partially paid, pending

Analysis

- Total number of payment records -> paymentStatus
- Results provide insight into the hospital's overall billing and payment performance.
- Detect financial information for the hospital- aware of problems that indicate insurance & billing

5 Best Queries: Query 4

Query 4 – Finding if the appointment time does not fall within the doctor's availability

- Joins appointments with doctor availability records.
- Identifies appointments scheduled outside the doctor’s available date or time range.
- Counts how many such scheduling conflicts exist.

```sql
-- Finding if the appointment time does not fall within the doctor's availability.
SELECT
COUNT(*) AS AppointmentsWithDoctorUnavailable
FROM HealthcareDB.Appointment a
JOIN HealthcareDB.DoctorAvailability da
ON a.DoctorAvailabilityID = da.DoctorAvailabilityID
WHERE
a.AppointmentTime NOT BETWEEN da.DoctorStartTime AND da.DoctorEndTime
OR a.AppointmentDate != da.DoctorDate;
```

Row	AppointmentsWithDoctorUnavailable
1	0

5 Best Queries: Query 5

Q. Which medical centers have the highest dispensing capacity, measured by how many pharmacies they own?

- This query identifies which medical centers have the largest pharmacy footprint.
- It joins medical centers with their associated pharmacies and counts how many pharmacies each center operates.
- A higher number of pharmacies can indicate higher potential patient traffic and prescription volume, since these centers can serve patients across more pharmacy locations.

Query results

Job information | Results | Visualization | JSON | Executi

Row	MedicalCenterID	MedicalCenterName	TotalPharmacies
1	2	Downtown Pediatric Clinic	1
2	5	Orthopedic Care Center	1
3	3	Radiology Imaging Center	1
4	1	Springfield General Hospital	1
5	4	NeuroHealth Institute	1

```
SELECT
  mc.MedicalCenterID,
  mc.MedicalCenterName,
  COUNT(ph.PharmacyID) AS TotalPharmacies
FROM HealthcareDB.MedicalCenter AS mc
LEFT JOIN HealthcareDB.Pharmacy AS ph
  ON mc.MedicalCenterID = ph.MedicalCenterID
GROUP BY
  mc.MedicalCenterID,
  mc.MedicalCenterName
ORDER BY
  TotalPharmacies DESC;
```

Thank you

Any Questions?